



Bladder and Other Urothelial Cancers Screening (PDQ®)–Health Professional Version

Overview

Note: The Overview section summarizes the published evidence on this topic. The rest of the summary describes the evidence in more detail.

Other PDQ summaries on [Bladder Cancer Treatment](#) and [Levels of Evidence for Cancer Screening and Prevention Studies](#) are also available.

Benefits

There is inadequate evidence to determine whether screening for bladder and other urothelial cancers has an impact on mortality.

Description of the Evidence

Study Design: There are no studies that directly address this question.

Internal Validity: Not applicable (N/A).

Consistency: N/A.

Magnitude of Effects on Health Outcomes: N/A.

External Validity: N/A.

Harms

Based on fair evidence, screening for bladder and other urothelial cancers would result in unnecessary diagnostic procedures with attendant morbidity.

Description of the Evidence

Study Design: Opinions of respected authorities based on clinical experience, descriptive studies, or reports of expert committees.

Internal Validity: N/A.

Consistency: N/A.

Magnitude of Effects on Health Outcomes: Good evidence for rare harms.

External Validity: N/A.

Incidence and Mortality

Bladder cancer is the fourth most commonly diagnosed malignancy in men in the United States. The incidence is about three to four times higher in men than in women and about two times higher in White men than in men of other ethnic and racial groups in the United States. The incidence of bladder cancer also increases with age.[1,2] It is estimated that 84,870 new cases of bladder cancer are expected to occur in the United States in 2025.[1]

Annual incidence rates for bladder cancer have been relatively stable from 1975 to 2021, ranging from 18.6 to 19.1 (per 100,000).[2] It is estimated that 17,420 Americans will die of bladder cancer in 2025.[1]

Age-adjusted mortality from bladder cancer decreased in all races and sexes between 1975 and 2022.[2] From 2013 to 2022, bladder cancer mortality decreased by about 1% per year.[1] These changes may reflect earlier diagnosis, better therapy, less exposure to carcinogens, or some combination of these factors.

More than 90% of cancers in the bladder are transitional cell carcinomas, also called urothelial cancer. Urothelial cancer can also rarely develop in the lining of the renal pelvis, ureter, prostate, and urethra. Other important histological types include squamous cell carcinoma and adenocarcinoma. Adenocarcinomas account for less than 2% of primary bladder cancers, including metastases from the rectum, stomach, endometrium, breast, prostate, and ovary.[3]

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Risk Factors

There are no definitive studies on the prevention of bladder or other urothelial cancers. Reduction in environmental and occupational exposures would presumably reduce urothelial cancer risk. Differences in age, sex, race, and geographic distribution may reflect differences in environmental and occupational exposure to possible toxicants. Relevant exposures include chemical exposures; cigarette smoking; infection with bacteria, parasitic fungi, or viruses; and treatment with certain chemotherapeutic agents. A positive family history of bladder cancer has also been associated with an increased risk of bladder cancer.[1,2]

Smoking

Several populations with a variety of exposures appear to be at higher risk of developing bladder cancer. By far, the greatest known environmental risk factor in the general population is tobacco, especially cigarette smoking.[1] Individuals who smoke have a fourfold to sevenfold increased risk of developing bladder cancer than individuals who have never smoked.[3-5] Risk is reduced with cessation of smoking, but a relatively small decrease in incidence is seen for the first 5 to 7 years after cessation. Even after 10 years, the risk of an individual developing bladder cancer is still almost twice that of an individual who has never smoked.

Among the chemicals implicated in smoking-induced bladder cancer are aminobiphenyl and its metabolites.[1] It is possible that inherited and inducible enzymes are important in the activation and detoxification of aminobiphenyls and other putative bladder carcinogens. These enzymes include N-acetyltransferase 2 (NAT2),[6] cytochrome P450 1A2 (CYT 1A2),[7] and glutathione S-transferase M 1.[8] Several studies have indicated that specific genotypes and phenotypes of these enzymes and their activities, particularly in the liver and urothelium, are associated with susceptibility to smoking-induced bladder cancer and bladder cancer induced by other aryl amines, particularly in industrially exposed populations.[7,9-12] Not all of these studies, however, have been well controlled for active or former smoking histories.

Environmental and Occupational Exposure to Certain Chemicals

A variety of industrial exposures have also been implicated as risk factors for developing bladder cancer, primarily aromatic amines, such as 2-naphthylamine, beta-naphthylamine, or 4-chloro-o-toluidine, present in the production of dyes and benzidine and its derivatives;[1] possibly chlorinated aliphatic hydrocarbons;[13] chlorination by-products in treated water; [14,15] aluminum production (polycyclic aromatic hydrocarbons, fluorides);[1] and certain aldehydes.[16]

Occupations reported to be associated with an increased risk of bladder cancer include those that involve processing paint, dye, metal, and petroleum products.[1,17]

It is estimated that 5% to 15% of patients in the United States who eventually die of bladder cancer have strong exposure histories to the above-named environmental factors (other than

smoking).[18]

Chinese Herbs

The use of contaminated Chinese herbs is also reported to be a risk factor. The prime carcinogen in these herbs appears to be aristolochic acid (AA) extracted from species of *Aristolochia*. [19] Because of the diversity of Chinese herbal regimens used in addition to AA, other unidentified phytotoxins may also play a role. [20] The chronic nephropathy associated with ingestion of herbs contaminated with *A. fangchi* has been linked to urothelial carcinoma of the renal pelvis and ureter. Herbs with *A. fangchi* are banned from Belgium, Canada, Australia, and Germany but are still available in the United States. [21]

Exposure to Arsenic

Ingestion of large quantities of arsenic in well water has also been associated with numerous malignancies, including transitional cell carcinoma (TCC) of the bladder. [1,22,23] Similar endemic pockets of bladder cancer are found in other regions with high arsenic concentrations in drinking water. [22,23] In South Taiwan, arsenic blackfoot disease is endemic.

Exposure to inorganic arsenic compounds, such as gallium arsenide, is also associated with an increased risk of bladder cancer.

Treatment With Cyclophosphamide or Ifosfamide

Exposure to the cancer chemotherapy agent cyclophosphamide [24,25] and perhaps other alkylating agents, such as ifosfamide (although the use of mesna in conjunction with these agents may reduce the incidence), is associated with an increased risk of bladder cancer. [24]

Pelvic Radiation Therapy

Pelvic radiation therapy for other malignancies, such as prostate cancer, [26,27] uterine cancer, [28] and cervical cancer, is also associated with an increased risk of bladder cancer. [29]

Genetic Mutations

Specific genetic mutations associated with bladder cancer include the following: [30-32]

- *HRAS* mutation (Costello syndrome or Faciocutaneoskeletal syndrome).
- *RB1* mutation.
- *PTEN/MMAC1* mutation (Cowden syndrome).
- *NAT2* slow acetylator phenotype.
- *GSTM1* null phenotype.

Other Risk Factors

Additional risk factors associated with more aggressive forms of bladder cancer include neuropathic bladder and associated indwelling catheters [33,34] and *Schistosoma haematobium* bladder infections (Bilharzial bladder cancer).[35]

Urothelial tumors other than TCC include adenocarcinoma, squamous cell carcinoma, and metastatic adenocarcinoma. Risks of squamous cell tumors in the bladder include indwelling catheters [36,37] and *S. haematobium* cystitis.

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Inheritance

Although occasional familial clusters have been anecdotally reported [1-3] and bladder cancer (as well as upper urinary tract transitional cell carcinomas) is part of the Lynch syndrome II,[4] there is no evidence that tendencies toward developing bladder cancer are inherited.[5]

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Clinical Presentation

Seventy percent of patients with bladder cancer have superficial disease at presentation.[1] Hematuria is the most common presenting sign, occurring in about 90% of cases. Hematuria may be intermittent, so a urinalysis without red blood cells does not exclude a diagnosis of urothelial cancer. In patients with macroscopic hematuria, the reported rates of bladder cancer range from 13% to 34.5%.[2-4] Other presenting symptoms include dysuria, urinary frequency or urgency, and less commonly, flank pain secondary to obstruction, and pain from pelvic invasion or bone metastases. Diagnosis and staging usually begin with cystoscopy. Full evaluation of the upper and lower urinary tract is required.[5]

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Histopathology

More than 90% of bladder cancers diagnosed in the United States are pure transitional cell carcinomas (TCCs) or TCCs mixed with other histologies, primarily squamous cell carcinomas (SCCs), adenocarcinomas, or both. An additional 3% to 4% are pure SCCs, which are approximately twice as likely to occur in women as in men. SCCs also represent a greater proportion of bladder cancers occurring in individuals who have *S. haematobium* infections of the bladder or who have histories of long-term indwelling urinary catheters, bladder stones, or recurrent bladder infections.[\[1-3\]](#)

Both the grade and stage at diagnosis of TCC have extremely important prognostic and therapeutic implications. Nontransitional cell histologies, however, all behave very aggressively and are less responsive to treatments other than extirpative surgery.[\[4\]](#) The prognosis of patients and the choice of treatments depend on the aggressiveness and grade of the tumor.

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Grade and Stage of Newly Diagnosed Bladder Cancer in an Unscreened Population

The critical nature of the histological grade and stage of index lesions for individual prognosis and management decisions has been well recognized for many years. In a study that attempted to evaluate grade and stage in newly diagnosed bladder tumors in a population-based setting, 89% of all newly diagnosed bladder cancers in men aged 50 years and older reported to the state of Wisconsin tumor registry in 1988 had blocks and slides reviewed by a single pathologist who did not know the original diagnosis.[1] Fifty-seven percent of specimens were grade 1 or 2, stage Ta or T1 transitional cell carcinomas (TCCs); 19% were grade 3, stage Ta or T1 (or Tis) TCCs; and 24% were muscularis propria invading or deeper (stage T2+), almost all of which were grade 3 lesions or of nontransitional cell histologies. Because of Wisconsin's small population of Black males aged 50 years and older (fewer than 3% of all bladder cancers occurred in non-White individuals),[2] differences in grade and stage at presentation between Black and White individuals could not be determined. Similarly, this study did not look at females or at males younger than 50 years. Because of variability in histological interpretations of bladder cancers recorded by tumor registries,[3,4] the presenting grade and stage of this malignancy in Wisconsin is known only for males aged 50 years and older.

Almost all bladder malignancies originate on the uroepithelial surface. Most patients who die of bladder cancer do so from metastatic disease; treatment for metastatic bladder cancer is rarely, if ever, curative.[5] The overwhelming majority of patients with metastases have concomitant or prior muscularis propria (stage T2+) invading lesions.[6] Seventy percent to 90% of patients with muscularis propria invading bladder cancer present with this diagnosis; however,[7,8] they do not come from the much larger pool of patients with recurring superficial TCCs. The goal of screening is the early detection of bladder cancer that is destined to become muscle invading. Although one study reported that approximately 30% of patients with superficial TCC followed for 20 years eventually die of this disease,[9] these data remain unconfirmed, are at odds with other reports,[10] and may reflect outmoded patterns of diagnosis, classification, and management.

Because bladder cancer is almost never incidentally found at autopsy, the preclinical duration in which it has not yet caused symptoms, but in which it can be detected by cystoscopy, is probably brief. This rapid growth rate is supported by clinical experience [11] and implies that screening would have to be performed at frequent intervals.

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Screening Methods

Cystoscopy and Cytology

The use of cystoscopies and bladder wash/urinary cytologic examinations has proven quite successful in the surveillance and management of patients with previously treated bladder cancers.^[1] These means are not practical in individuals without a history of bladder cancer because of expense and morbidity.

Hematuria

Although hematuria is the most common presenting sign of bladder cancer, most individuals with hematuria do not have bladder cancer. In the general population, the prevalence of asymptomatic gross hematuria is about 2.5%, while the prevalence of asymptomatic microhematuria is about 13%.[2] In a recent prospective analysis of patients attending a hematuria clinic in the United Kingdom, 183 of the 948 patients (19.2%) with gross hematuria were found to have bladder cancer on cystoscopy.[3] However, only 47 of the 982 patients (4.8%) with microhematuria were found to have bladder cancer.

One-Time Hematuria Testing

Two groups have reported on the use of testing a single urine specimen for blood to detect urologic malignancies, serious urinary tract diseases, and bladder cancers. Both studies were performed retrospectively to ascertain information from patients who were seen at a large multispecialty clinic [2] or who subscribed to a large health maintenance organization (HMO) and were tested in a multiphasic screening.[4] Because of the retrospective nature of each study, neither was designed to specifically look for bladder cancer detection or to focus on the population at highest risk (men aged 50 years and older). Both studies concluded that single hematuria testing was not effective in diagnosing bladder cancer. A longer follow-up of the HMO study indicated that individuals with microhematuria were at a higher risk of subsequent development of muscle-invading bladder cancer, with a latency of 3.5 to 14.5 years.[5] There is insufficient evidence to indicate that single hematuria testing is effective in screening for bladder cancer, and there is no evidence that single hematuria testing results in reduced mortality from the disease.

Repetitive Hematuria Testing

Two studies using Ames Hemastix, a chemical reagent strip for hemoglobin that correlates with microscopic urinalysis in detecting hematuria,[6] were conducted in geographically defined (Madison, Wisconsin and Leeds, England) populations of middle-aged and older men using repetitive home reagent strip testing. In each program, patients were solicited from patient care registries. Men with histories of previous urologic malignancies, or known causes of hematuria, or who were noncompliant were eliminated. In the four studies performed (one pilot study and one larger study at each site), 45% to 55% of solicited individuals took part. In these studies, 1.2% to 1.3% of all participants were found to have bladder cancer (all transitional cell carcinomas [TCCs]). Only 1 of the 21 patients in the first study [6-8] and none of the 26 detected in the second study had stage T2 or greater malignancy. As a limitation of repetitive hematuria screening in a general population of men aged 50 years and older, more than 90% of individuals with positive tests upon initial workup were found not to have bladder cancer.[6] In the Wisconsin hematuria screening studies, all patients who were hematuria positive with negative workups or who were found to have no serious disease were followed

for at least 24 months, with no findings of developing bladder cancer. Similarly, at least 18 months after their last testing, no screening participant (with or without hematuria) had died of bladder cancer. In the 14-year follow-up of this screening cohort, no participant with bladder cancer detected by hematuria screening had died of bladder cancer, while two (0.85%) with hematuria and a negative workup developed the disease 6.7 and 11.4 years after their negative workup. The same proportion of participants without hematuria during screening were diagnosed with bladder cancer (0.93%), none within 1 year after their last testing date.[9] It is possible that longer follow-up is necessary to prove that these participants did not have bladder cancer;[5] however, such studies are not available. The relatively low positive predictive value of repetitive hematuria testing (7.6% for bladder cancer and 11.6% for all malignancies) [6,8,10] raises questions about the practicality of this mode of screening.

Other Possible Screening Modalities

The accuracy of voided urine cytology in detecting bladder cancer has been evaluated primarily in patients with histories of bladder cancer who are undergoing cystoscopic surveillance or as a routine test performed in all patients attending a large urology office in a multispecialty clinic. In the studies of patients with histories of bladder cancer, voided urinary cytology was effective in diagnosing 20% to 40% of grade 1 TCCs, 20% to 50% of grade 2 malignancies, and 60% to 80% of grade 3/Tis cancers.[11,12] Although such studies were not performed in patients without either hematuria or histories of recurrent bladder tumors, a major concern for screening purposes is the lack of sensitivity for well-differentiated and moderately differentiated TCCs and the large proportion of specimens in which an insufficient number of cells were present for any cytologic diagnosis to be offered. Although false-positive results were exceedingly rare, the lack of sensitivity even in this highly suspect population make voided urine cytology an inappropriate test for screening the general population. No studies have looked at outcome of cytologic screening on disease-related mortality in a non-industrially exposed population. Outcomes of patients screened at the urology clinic are also not available. [11]

The outcomes of men diagnosed with bladder cancer through a hematuria home screening program using a chemical reagent strip were compared with a statewide population-based sample of 87% of all men aged 50 years and older from the Wisconsin tumor registry.[8] Histological sections were blindly reviewed, and similar proportions of low-grade superficial versus high-grade or invasive cases were found; the proportion of late-stage (T2 or higher) disease was lower in the screened patients. At 14 years, 20.4% of tumor registry patients had died of bladder cancer (including 50% of those with muscle-invading grade 3 lesions); however, at 14 years of follow-up, no participant with bladder cancer detected by screening had died of bladder cancer.[13] Whether these differences resulted from some combination of lead-time effect, overdiagnosis, and selection biases, a real screening effect cannot be determined.

The measurement of a variety of molecules and cellular elements screened in urine has been proposed, and in some cases marketed, to monitor previously diagnosed bladder cancer patients. However, the specificity and sensitivity of these markers have not been assessed in a screening setting in a general population, but several such studies are under way.

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Special Populations

In populations at particularly high risk of developing bladder cancer (other than those with histories of bladder cancer), few screening studies that have assessed bladder cancer mortality have been published.[\[1-4\]](#) A study of annual cytology in aluminum workers exposed to coal tar pitch in Quebec showed a nearly 40% reduction in bladder cancer case-fatality 6 years after diagnosis, compared with a historical control group of workers from the same plants who were not screened;[\[5\]](#) the difference, however, was not statistically significant. Awareness of an adverse outcome in the unscreened predecessors may have influenced participation in the program and workers' awareness of symptoms, the willingness of workers and physicians to initiate diagnostic investigations based on signs and symptoms, and the compliance of workers with medical recommendations for evaluation and treatment. The brief duration of follow-up in the screened group may have artifactually improved the outcome.

No randomized controlled bladder cancer screening trials have been conducted in environmentally or industrially exposed cohorts. Completed studies have usually not had comparable control groups, have not been of sufficient sample size to show an effect on outcome, and have been of insufficient length to show a mortality benefit (or lack thereof) for the modality or modalities being assessed.[\[2,3\]](#) One study described the usefulness of measuring three biomarkers in voided urine for risk assessment and cancer detection in a large cohort of Chinese workers at increased risk of bladder cancer.[\[6\]](#) The workers were individually stratified, screened, monitored, and diagnosed on the basis of predefined molecular biomarker profiles. These techniques remain investigational.

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Latest Updates to This Summary (04/08/2025)

The PDQ cancer information summaries are reviewed regularly and updated as new information becomes available. This section describes the latest changes made to this summary as of the date above.

Incidence and Mortality

Updated [statistics](#) with estimated new cases for 2025 (cited American Cancer Society as reference 1).

Updated [statistics](#) with estimated deaths for 2025.

Revised [text](#) to state that from 2013 to 2022, urinary bladder cancer mortality decreased by about 1% per year.

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About This PDQ Summary

Purpose of This Summary

This PDQ cancer information summary for health professionals provides comprehensive, peer-reviewed, evidence-based information about the screening of bladder and other urothelial cancers. It is intended as a resource to inform and assist clinicians in the care of their patients. It does not provide formal guidelines or recommendations for making health care decisions.

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